Power & link

File: main_power.kicad_sch

MCU, flash & debug

File: mcu.kicad_sch

OLED display & 13 V boost

File: display.kicad_sch

LED ring

File: leds.kicad_sch

Sensors & sounder

File: sensors.kicad_sch

Detent actuator & home sensor

File: detent.kicad_sch

dial-me-spinner rev A — generated by hardware/design/build.py from hardware/design/sheets/*.py.
Do not hand-edit the .kicad_sch files: change the sheet modules and rebuild.
Architecture of record: hardware/design/SPEC.md

dial-me-spinner

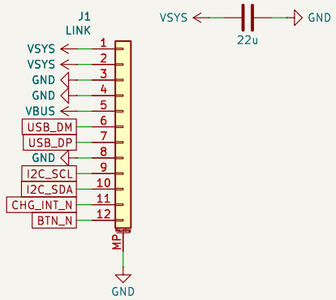
Sheet: /
File: dial-me-spinner-main.kicad_sch

Title: dial-me-spinner main board

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/7

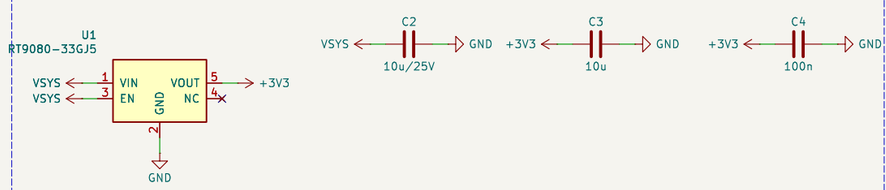
Board-to-board link (to base board through the spindle)

JST SH 12-way SIDE-entry on the bottom at r = 19.5, θ 330° (a vertical header would hit the rotor hub); the harness lies flat under the board into the spindle bore. Identical pinout on the base board (SPEC §2)



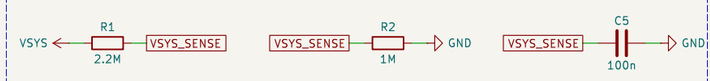
3V3 always-on LDO

nRF VDD+VDDH, flash, sensors, OLED logic; Iq 2 uA



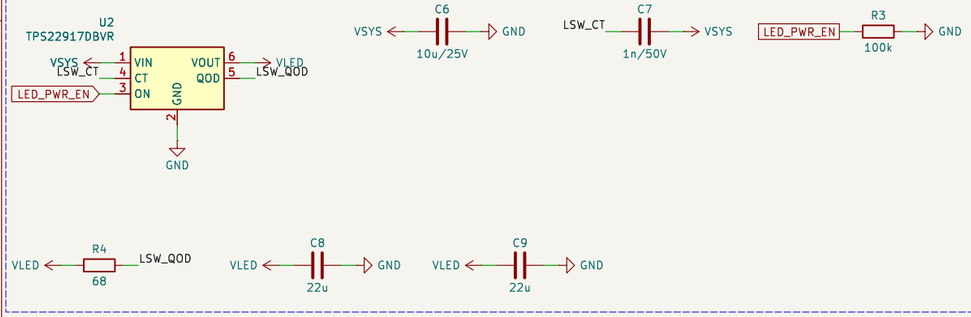
VSYS sense

2.2M/1M always-on divider (~1.3 uA) into AIN0; 100 nF holds the sample



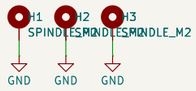
LED rail load switch

VSYS -> VLED; CT 1 nF to VIN (TPS22917 pin table) ~4 ms rise; QOD discharge through R4



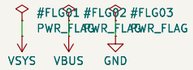
Spindle clamp holes (GND)

Plated M2 holes for the stainless spindle flange: grounds spindle -> 608 races -> ring, giving the touchable 316L flywheel an ESD bleed path instead of arcing to the encoder (review SCH-2). Bond the ring to the bearing outer race with conductive epoxy (docs/mechanical.md).

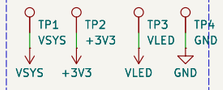


Power flags

VSYS, VBUS and GND enter this board only through the passive link connector J1



Test points



dial-me-spinner

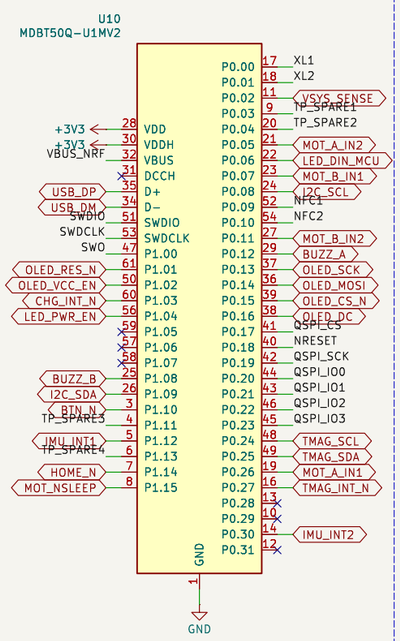
Sheet: /Power & link/
File: main_power.kicad_sch

Title: Power & link

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 2/7

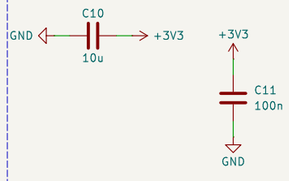
nRF52840 module MDBT50Q-U1MV2

Normal-voltage mode: VDD + VDDH on +3V3, DCCH n/c, REG1 DC/DC on (its inductor is inside the module). u.FL on module -> Molex 146153-0100 flex antenna. Pin map = nets.MCU_PINS (SPEC §4). UICR: P0.18 = nRESET; NFCPINS stays in NFC mode (P0.09/P0.10 = NFC1/NFC2 antenna). Symbol: stock MDBT50Q-U1MV2 pins, regrouped by port.



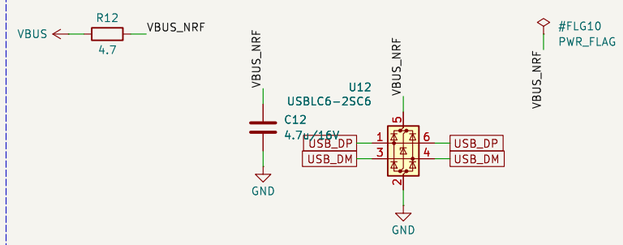
Module VDD/VDDH decoupling

At pads 28/30, < 2 mm (Raytac ref. circuit 8.3: 10 uF)



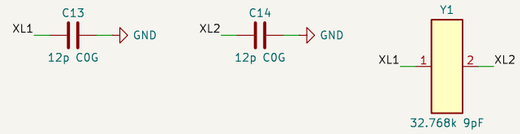
USB VBUS (module pad 32) and second-stage USB ESD

R12 in series (nRF52840 PS 7.3 config 1): a legacy-cable hot plug rings the link VBUS to 6.5-8.4 V, above the 5.8 V VBUS abs max; with 4.7 R (Nordic's value) R12 x C12 filters the ring and the pin sees the settled VBUS, 4.49-5.49 V for a 4.5-5.5 V source (hardware/sim test_hotplug_nrf_vbus_abs_max). DC drop ~11 mV at the 2.4 mA VBUS current. C12 on the module side; total sink C < 10 uF. U12 USBLC6 at the module: IEC 8 kV on D+/D- otherwise drives 2.6-3.7 A into the nRF pads through the ~145 nH link (HBM 2 kV class = 1.33 A). U12 as close as the module courtyard allows (3.5 mm pad to pad to D-); test_usb_input reads the placed distance.



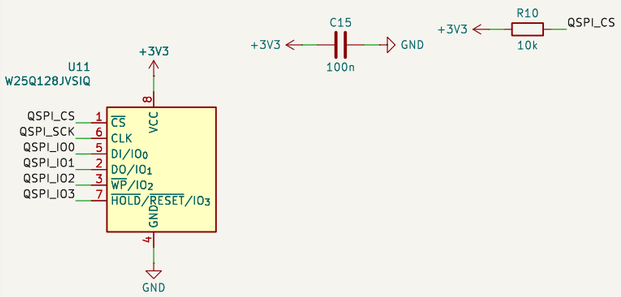
32.768 kHz LFXO

*SC-20S CL 9 pF; Ccap = 2*CL - Cpin - Cpcb ~ 12 pF C0G. R1 <= 90k (nRF RS_LFXO max 100k); DLmax 0.5 uW = nRF PD_LFXO. At pads 17/18, traces < 5 mm, symmetric, GND guard; no copper under Y1.*



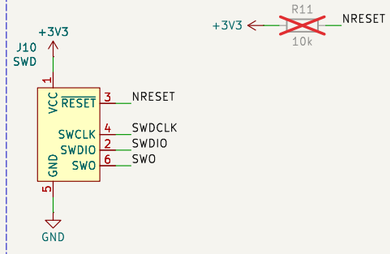
QSPI NOR flash (128 Mbit)

nRF52840-DK QSPI pins (pads 41-46): traces < 20 mm, matched within 2 mm. -IQ suffix: QE=1 fixed, /WP and /HOLD disabled, so IO2/IO3 need no pull-ups. CS pulled up so the flash stays deselected in reset. Use deep power-down when idle.



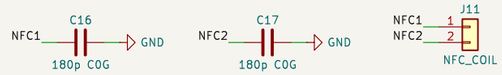
SWD debug (Tag-Connect TC2030-NL)

Footprint only (no BOM part), TOP side under the OLED glass (SPEC section 8). Keep the 3 NPTH holes + cable-head area clear. No series parts on SWD. NRESET: internal pull-up, R11 is a DNP option.



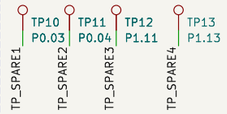
NFC tag antenna (nRF52840 NFCT, NFC-A Type 2/4)

*Coil between NFC1 and NFC2: its flex tail solders to pads J11 (coil in the top cover). Tune: Ctune = 2/((2*pi*13.56 MHz)^2 * L_coil) - Cp - Cint (Cint ~ 4 pF, PS 6.13): 180 pF C0G each suits L ~ 1.4-1.5 uH; retune on prototype. Route NFC1/NFC2 as a pair, away from the TMAG and the LED data line.*



Spare GPIO test pads

Value = module GPIO (P0.03/P0.04 = AIN1/AIN2)



dial-me-spinner

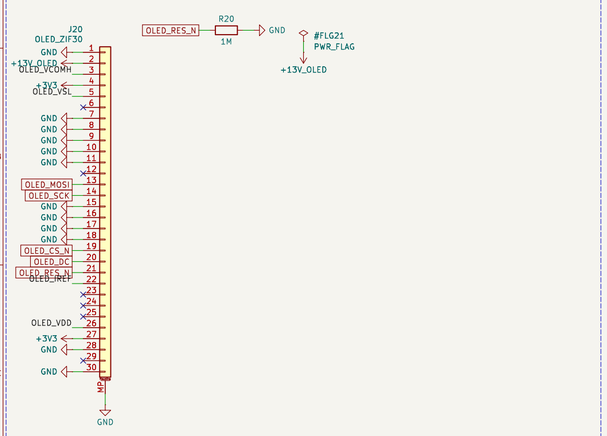
Sheet: /MCU, flash & debug/
File: mcu.kicad_sch

Title: MCU, flash & debug

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 3/7

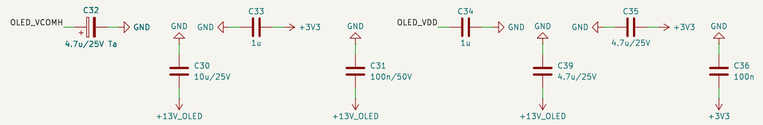
OLED ZIF J20: NHD-1.5-128128G panel, 30-pin 0.5 mm FPC

Pin map from Newhaven NHD-1.5-128128G datasheet, Pin Description + Interface Selection (p.4), Wiring Diagram (p.5):
https://newhavendisplay.com/content/specs/NHD-1.5-128128G.pdf (Orient AOC128128A0-1.50W is pin-identical)
1 NC(GND)=GND | 2 VCC=+13V_OLED | 3 VCOMH | 4 VDDIO=+3V3 | 5 VSL | 6 NC | 7-11 D7..D3=GND
12 D2 open (SPI) | 13 D1=SDIN | 14 D0=SCLK | 15 E/RD#=GND | 16 R/W#=GND | 17 BS0=GND | 18 BS1=GND
19 CS# | 20 D/C# | 21 RES# | 22 IREF | 23 GPIO1, 24 GPIO0, 25 NC: open | 26 VDD (int. reg.)
27 VCI=+3V3 | 28 VSS | 29 NC | 30 NC(GND)=GND. BS1:BS0 = 00 -> 4-wire SPI, write-only, mode 0, <= 20 MHz.
Bottom-contact ZIF: FPC folds 180 deg behind the glass (NHD's top-contact advice is for unfolded entry).
RES# 1M pull-down keeps the panel in reset while the MCU is in reset. Mates with the panel (not assembled).
FPC fits NHD W 15.5+/-0.2 / t 0.3+/-0.03 / stiffener 4+/-0.5 vs JUSHUO slot 15.60+/-0.05, recommended FPC
15.50+/-0.05 / t 0.30+/-0.01 / stiffener 6.0: check insertion, latch and retention with real panel samples
(with the 1:1 contact-side mock-up) before PCBA.



Panel supply capacitors (at the ZIF pins)

NHD wiring diagram: VCC C5 0.1u + C6 10u (C31, C30). VCOMH C7 4.7u/25V tantalum (C32).
VDDIO C3 1u (C33), VDD C4 1u (C34), VCI C1 0.1u + C2 4.7u (C36, C35).
C39 = boost output cap on +13V_OLED: place it at D21 / U20 GND, not at the ZIF.
No bleeders or QOD switches on +3V3 / +13V_OLED (VCC must float when off).



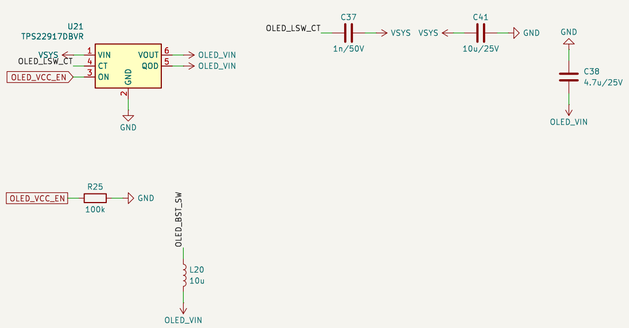
IREF and VSL networks

IREF: R = (VCC - 6 V) / 12.5 uA = 560k at 13 V (recompute if VCC changes).
VSL: 50 ohm 1/4 W + -1.4 V: BAV99 as 2 series diodes (pin 1 A1, pin 3 mid NC, pin 2 K2).



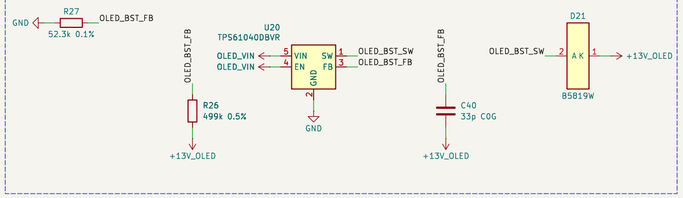
Boost input: load switch VSYS -> OLED_VIN, CIN, inductor

OLED_VCC_EN low (also through MCU reset, R25) -> U21 off: VCC floats, boost draws 0 uA.
Without it a disabled boost still passes VSYS - Vf to VCC through L20 and D21.
U21 TPS22917: CT 1 nF from CT to VIN (C37) -> ~4 ms ramp. Inrush is set by the TPS61040 soft start
(it starts at its 1.5 V UVLO while U21 ramps): ~0.3 A for ~2 ms, 0.4 A peak.
QOD tied to OLED_VIN: when off, OLED_VIN is held at ~0 V (150 R) so the boost (EN = VIN) stays off:
otherwise D21 leakage from the panel's VCI->VCC ESD path floats it up and the boost hiccups in sleep.
(Replaces the AO3401A/2N7002 disconnect, which had no real dV/dt control - review SCH-1.)
LAYOUT: L20 and the SW node >= 10 mm from the TMAG5273 and the magnet track.



13 V boost: TPS61040, rectifier, FB divider

VOUT = 1.233 V x (1 + 499k / 52.3k) = 13.00 V (0.5 % / 0.1 % divider): CFF C40 33 pF across R26 (tune 22-100 pF).
EN tied to OLED_VIN: the input disconnect does the switching. L 10 uH, Isat 0.72 A rated (0.92 A typ; peak need 0.49 A).
LAYOUT: tight CIN-L-SW-GND and SW-D-COUT-GND loops; R26/R27/C40 at the FB pin, away from SW.



dial-me-spinner
Sheet: /OLED display & 13 V boost/
File: display.kicad_sch

Title: OLED display & 13 V boost

Size: A2	Date: 2026-10-19	Rev: A
KiCad E.D.A. 9.0.9		Id: 4/7

48× WS2812C-2020-V6 (5 mA/ch) on VLED = TPS22917 load-switch output (main_power sheet, 2× 22 µF bulk there).
Data: nRF P0.06 (I2S SDOUT, HIGH drive) → LED_DIN_MCU → R40 220R → LED_DIN_R (R41) → R43 0R → LED_DIN → … → D148 (TP40).
D101 = pixel 0 at the mechanical index; pixels run in chain order around the ring. 3.3 V logic: DNP SN74LV1T34 option (U43/R43) if the margin is short.
WS2812B-2020-V6 (C529174JH, 12 mA/ch) is a footprint/pinout drop-in; firmware must then cap peak current.

LED data input

220R series (bounds DIN clamp current if P0.06 is high while VLED is off); place within 3 mm of the nRF pin (P0.06).
100k pull-down R41 on LED_DIN_R holds DIN low while P0.06 is
high-Z (reset/DFU). FIRMWARE: send an all-zero frame and
drive P0.06 low (glitch) before deasserting
LED_PWR_EN; after raising it, wait >= 10 ms with P0.06 low.
Never drive LED_DIN_MCU high while VLED is off:
QDD holds VLED at 0 V and VI max = VDD + 0.7 V, so DIN
would back-power the ring through its ESD diode.

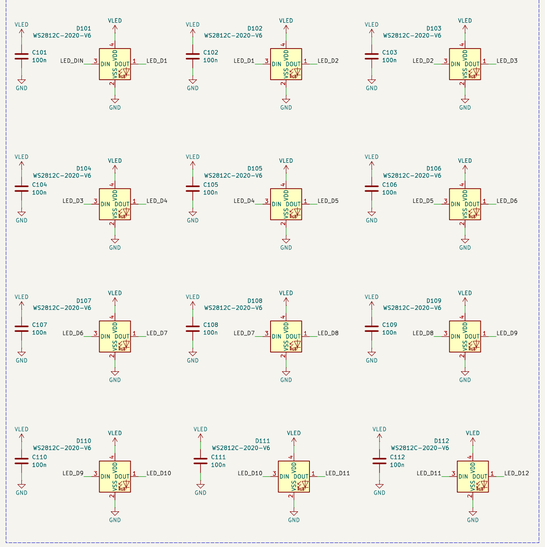
Optional VLED-referenced level shifter (DNP)

Default: R43 0R fitted, U43 not fitted (3.3 V driver, DC margin over 0.6 VDD +427 mV typical, +41 mV
datasheet bound at VSYS 4.59 V, so SYS_REG stays 4.4 V and the D101–D148 must be V6 parts).
If the bench test at VSYS = 4.6 V shows bit errors: remove R43, fit U43 (powered from VLED,
so it is unpowered whenever the ring is off) and C43.



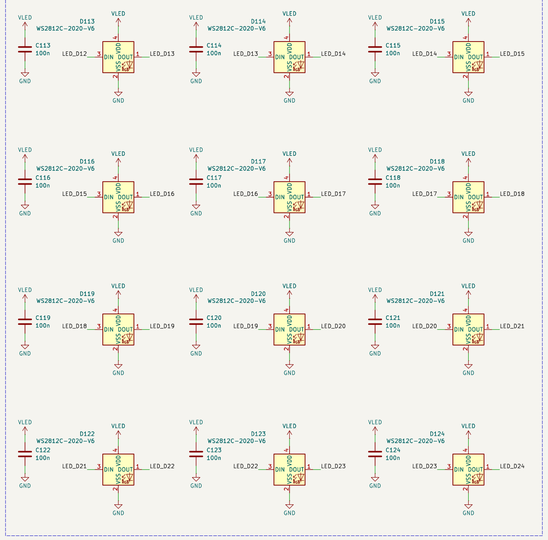
LED ring D101–D112 (pixels 0–11)

chain DOUT -> next DIN: C101–C112: one 100 nF 0402 per LED,
<= 1 mm from its VDD pad, sharing the VDD/GND vias



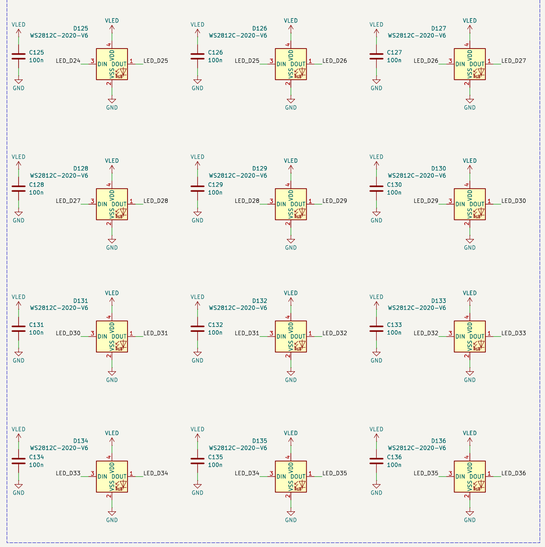
LED ring D113–D124 (pixels 12–23)

chain DOUT -> next DIN: C113–C124: one 100 nF 0402 per LED,
<= 1 mm from its VDD pad, sharing the VDD/GND vias



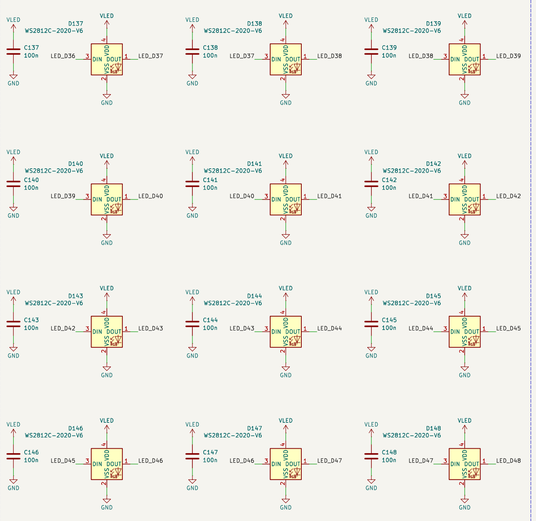
LED ring D125–D136 (pixels 24–35)

chain DOUT -> next DIN: C125–C136: one 100 nF 0402 per LED,
<= 1 mm from its VDD pad, sharing the VDD/GND vias



LED ring D137–D148 (pixels 36–47)

chain DOUT -> next DIN: C137–C148: one 100 nF 0402 per LED,
<= 1 mm from its VDD pad, sharing the VDD/GND vias



Chain end: D148.DOUT = LED_D48

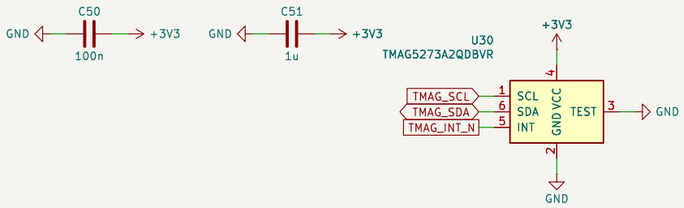
Test pad + 100k pull-down. R6 (0/DO) are
swappable; so DOUT must not float.
Never loop LED_D48 back to D101.



dial-me-spinner	
Sheet: /LED ring/	
File: leds.kicad_sch	
Title: LED ring	
Size: A1	Date: 2026-10-19
KiCad E.D.A. 9.0.9	Id: 5/7
	Rev: A

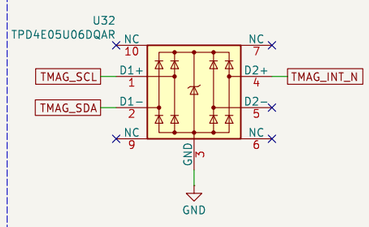
Rotation encoder: TMAG5273A2 3D Hall (TWIM0, addr 0x35)

BOTTOM side, theta = 270 deg, long axis radial (X tangential). Put the Hall element, not the package centre, on R_t = 29.0 mm: per TI Fig 6-2 it sits 0.40 mm from the body centre toward the pin-3/4 end and 0.12 mm toward the pin 1-3 side (footprint coords -0.12, +0.40 mm). Caps next to pin 4. No pours/inductors within the magnet-track keep-out.



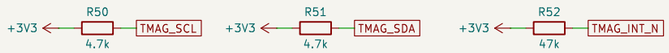
Encoder ESD (floating 316L ring 2.4 mm below)

The flywheel is a touchable, electrically floating metal ring passing under U30. U32 clamps TMAG_SCL/SDA/INT_N right at U30 (short GND return via next to U30).



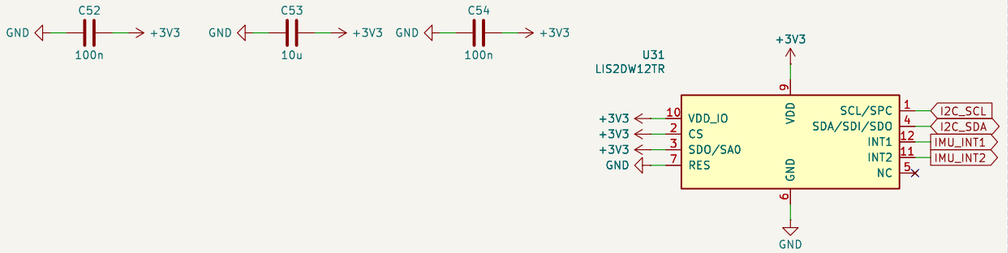
Encoder bus pull-ups (dedicated TWIM0)

4.7k SCL/SDA; 47k on open-drain INT (10 us pulses / latched)



Accelerometer: LIS2DW12 (system I2C, addr 0x19)

CS high = I2C mode. SA0 high = 0x19: never ground SA0 (its internal pull-up cannot be disabled). RES to GND. 100n + 10u at VDD (pin 9), 100n at VDD_IO (pin 10).



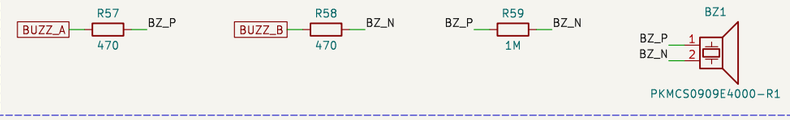
System I2C + link signal pull-ups (the only ones on the board)

System I2C goes to IMU + BQ25180 (0x6A, via link). CHG_INT_N / BTN_N are open drain on the base board.



Piezo sounder: Murata PKMCS0909E4000-R1 (magnet-free, 4 kHz, external drive)

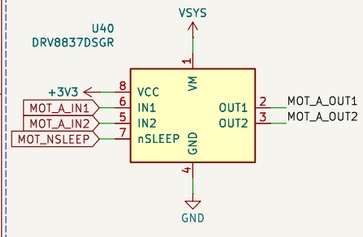
Complementary PWM on BUZZ_A/BUZZ_B (one nRF PWM instance): +/-3.3 Vo-p (max +/-12.5).
Murata land pattern: two 1.3 x 3.4 mm pads at x = +/-4.35 mm (custom footprint).
Sound exits via 2 mm slots in the bottom face at the non-terminal edges: keep them clear.

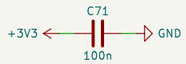


dial-me-spinner		
Sheet: /Sensors & sounder/		
File: sensors.kicad_sch		
Title: Sensors & sounder		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 6/7

Coil A driver — DRV8837 #1

IN1/IN2: 00 coast, 10 fwd, 01 rev, 11 brake
VM = VSYS (3.0–4.5 V), VCC = +3V3
Pin 4 + EP (pad 9) to the GND plane
100 nF at the VM pin and at the VCC pin
U40/U41 + caps: TOP side in theta 60–120, or
BOTTOM only outside theta 60–120 (keep-out)

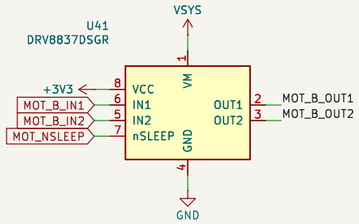


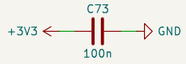


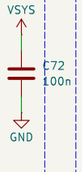


Coil B driver — DRV8837 #2

Same as #1. A 2-wire DC gearmotor uses
coil A only (U40, J40 pins 1–2); U41 then
idles with IN1/IN2 low

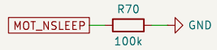






Shared VM bulk + nSLEEP pull-down

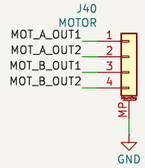
10 uF/25 V 0805 placed between the two VM pins
nSLEEP: 100k ext. pull-down (SPEC §4) in
parallel with the 100k internal one in each DRV
Sleep: IVM+IVCC <= 2x(95+25) nA = 0.24 uA max





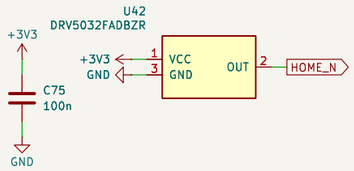
Motor connector (JST SH 4-way)

Side entry SM04B, ~2.9 mm tall: the lead
runs in-plane to the radial actuator
Bottom side: just outside the theta 60–120
keep-out, r < 27, entry facing the actuator
1 A+, 2 A−, 3 B+, 4 B−
2-wire DC gearmotor: pins 1–2 only
MP solder tabs to GND



Home (park) Hall switch

DRV5032FA omnipolar push-pull, always on
(+3V3, 1.6 uA typ avg). HOME_N low = detent
magnet parked (free mode): no pull-up needed
TOP side only, above park pos. r 21.5, theta 90
(bottom = keep-out, 0.5 mm to the magnet)



dial-me-spinner		
Sheet: /Detent actuator & home sensor/		
File: detent.kicad_sch		
Title: Detent actuator & home sensor		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 7/7